GPT-6.1 Sol

10MAA End of Year Revision Examination 2026

Marking Schedule

General comments

Accept equivalent methods and exact answers unless a particular form or accuracy is requested. Award method marks for valid working. For questions worth more than one mark, an unsupported correct answer earns the final-answer mark only. Do not penalise the same arithmetic error twice; allow follow-through where the subsequent method is valid. Penalise incorrect rounding only once across the paper. Money is given to two decimal places.

Algebraic Expressions & Equations • 34 marks

Question	Working and answer	Marks	Comments
1.	$8x - 12 = 3x + 23$ $5x = 35$ $x = 7$	2	1 for collecting terms. 1 for $x = 7$.
2.(a)	$9 - 4t \le 3t + 30$ $-7t \le 21$ $t \ge -3$	3	1 for collecting terms. 1 for boundary -3. 1 for reversing the inequality.
(b)	Closed circle at -3; ray to the right.	1	Allow follow-through from part (a).
3.	$3(-3)^2 - 2(-2)(5)$ $= 27 + 20 = 47$	2	1 for correct substitution. 1 for 47.
4.(a)	$\dfrac{18p^7q^3}{6p^2q} = 3p^5q^2$	2	1 for coefficient 3. 1 for both powers correct.
(b)	$(81k^8)^{\frac{3}{4}} = 81^{\frac{3}{4}}k^6 = 27k^6$	2	1 for 27. 1 for k^6.
5.(a)	$6x^2 - 54 = 6(x^2 - 9)$ $= 6(x - 3)(x + 3)$	2	1 for extracting 6. 1 for complete factorisation.
(b)	$\dfrac{5(x+1) - 2(x-2)}{(x-2)(x+1)}$ $= \dfrac{3x + 9}{(x-2)(x+1)}$	3	1 for common denominator. 1 for correct numerator expansion. 1 for simplification. Accept $3(x+3)$ in numerator.
6.	$T^2 = 5r + 4$ $r = \dfrac{T^2 - 4}{5}$	2	1 for squaring correctly. 1 for making r the subject.
7.	$a + 40b = 86$, $a + 70b = 122$ $30b = 36$, so $b = 1.20$ and $a = 38$ $a + 100b = 38 + 120 = 158$ **Cost = \$158.00**	3	1 for a valid pair of equations. 1 for a or b. 1 for \$158.00.
8.	$\dfrac{1}{2}(2x - 1)(x + 3) = 30$ $2x^2 + 5x - 63 = 0$ $(2x - 9)(x + 7) = 0$ $x = -7$ or $x = 4.5$ $\boldsymbol{x = 4.5}$ **cm**; reject -7 because it gives negative side lengths.	4	1 for area equation. 1 for quadratic equal to zero. 1 for both roots. 1 for 4.5 cm only, with a contextual reason for rejecting -7.

Algebraic Expressions & Equations continued			
Question	**Working and answer**	**Marks**	**Comments**
9.(a)	$\frac{60}{x}$ minutes	1	Correct expression.
(b)	$\frac{60}{x+10}$ minutes	1	Correct expression.
(c)	$\frac{60}{x} - \frac{60}{x+10} = 1$	2	1 for the difference of the two times in the correct order. 1 for equating to 1. Allow follow-through.
(d)	$60(x+10) - 60x = x(x+10)$ $x^2 + 10x - 600 = 0$ $(x+30)(x-20) = 0$ $x = -30$ or $x = 20$ **20 jobs per hour**. Reject -30 because a processing rate must be positive.	4	1 for eliminating denominators. 1 for the correct quadratic. 1 for solving it. 1 for 20 only, with a contextual reason for rejecting -30.

Statistics & Probability • 26 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.(a)	Mode $= 2$ projects.	1	Correct answer.
(b)	The 15th and 16th values are both 2. Median $= 2$ projects.	1	Correct answer.
(c)	$\text{Mean} = \frac{0(3) + 1(6) + 2(10) + 3(7) + 4(4)}{30}$ $= \frac{63}{30} = 2.1$ projects.	2	1 for weighted total or correct fraction. 1 for 2.1.
2.	$\text{Mean} = \frac{12(8) + 8(11) + 5(14)}{25}$ $= \frac{254}{25} = 10.16$ **10.2 seconds** to 3 significant figures.	3	1 for weighted totals. 1 for correct overall total and divisor. 1 for answer.
3.	Both $= 24 + 19 - (40 - 7) = 10$. Python only $= 14$; Rust only $= 9$; neither $= 7$. Exactly one $= 14 + 9 = 23$ students.	3	1 for intersection 10. 1 for all four diagram regions correct. 1 for 23.
4.(a)	$\frac{36}{100} = \frac{9}{25}$	1	Accept any equivalent probability.
(b)	$\frac{24}{30} = \frac{4}{5}$	1	Denominator must represent macOS computers.
(c)	$\frac{40 + 22 - 4}{100} = \frac{58}{100} = \frac{29}{50}$	1	The overlap must be counted once.
(d)	$\frac{12}{100} \times \frac{11}{99} = \frac{1}{75}$	2	1 for sampling without replacement, with both counts reduced for the second draw. 1 for final answer.

Statistics & Probability continued			
Question	**Working and answer**	**Marks**	**Comments**
5.(a)	Cumulative frequencies: $16, 48, 104, 144, 160$.	1	All five correct.
(b)	Plot $(0,0), (20,16), (40,48), (60,104),$ $(80,144), (100,160)$ and join in order.	2	1 for correct upper-boundary points. 1 for an increasing curve or joined line segments. Allow follow-through from (a).
(c)	Upper quartile is the time at cumulative frequency 120. $60 + \dfrac{120-104}{144-104} \times 20 = 68$ s.	1	Accept 66–70 s, or a consistent reading from the student's graph.
6.(a)	First check: Pass $\dfrac{3}{5}$; Fail $\dfrac{2}{5}$. After Pass: Pass $\dfrac{3}{4}$; Fail $\dfrac{1}{4}$. After Fail: Pass $\dfrac{1}{2}$; Fail $\dfrac{1}{2}$.	3	1 for each complete pair of branches.
(b)	$\dfrac{3}{5} \times \dfrac{1}{4} + \dfrac{2}{5} \times \dfrac{1}{2}$ $= \dfrac{3}{20} + \dfrac{1}{5} = \dfrac{7}{20}$	2	1 for a correct single-pass path. 1 for the sum $\dfrac{7}{20}$.
(c)	$P(\text{second pass}) = \dfrac{3}{5} \times \dfrac{3}{4} + \dfrac{2}{5} \times \dfrac{1}{2}$ $= \dfrac{9}{20} + \dfrac{4}{20} = \dfrac{13}{20} = 0.65$ $0.65 > 0.5$, so the claim is correct.	2	1 for $\dfrac{13}{20}$. 1 for a comparison with $\dfrac{1}{2}$ and a correct conclusion.

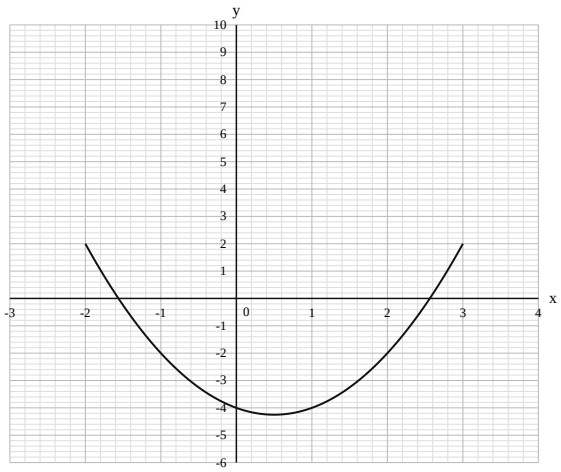
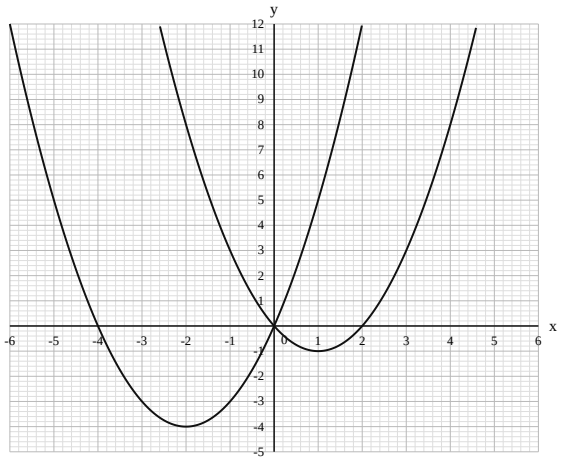
Trigonometry • 12 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.	$\tan 28° = \dfrac{h}{36}$ $h = 36 \tan 28° = 19.1415...$ **19.1 m**	2	1 for a correct trigonometric equation. 1 for answer.
2.(a)	$\cos \angle QPR = \dfrac{80^2 + 120^2 - 60^2}{2(80)(120)}$ $= \dfrac{43}{48}$ $\angle QPR = 26.3843...°$ **26.4°**	2	1 for cosine rule with correct substitution. 1 for angle.
(b)(i)	$60° + 26.3843...° = 86.3843...°$ **086.4°**	1	Allow follow-through from (a).
(ii)	$86.3843...° + 180° = 266.3843...°$ **266.4°**	1	Allow follow-through from (b)(i).
(c)	$120 \sin(86.3843...°) = 119.761...$ $\mathbf{1.20 \times 10^2}$ **km**	2	1 for $120 \sin(\text{bearing})$. 1 for answer. Accept 120 km.
(d)	$\dfrac{RS}{\sin 42°} = \dfrac{60}{\sin 73°}$ $RS = \dfrac{60 \sin 42°}{\sin 73°} = 41.9825...$ **42.0 km**	2	1 for correct sine rule. 1 for answer.
(e)	$A = \dfrac{1}{2}(80)(120)\sin(26.3843...°)$ $= 2133.0729...$ $\mathbf{2.13 \times 10^3 km^2}$	2	1 for correct area expression. 1 for answer. Use unrounded angles where possible.

Measurement • 14 marks

Question	Working and answer	Marks	Comments
1.	$0.48 \times 100^2 = 4800\text{cm}^2$	1	Correct answer.
2.	$A = 24(10) - 2\pi(3)^2 = 240 - 18\pi$ $= 183.451...$ **183 cm²** to the nearest cm².	3	1 for circle area using radius 3. 1 for subtracting both holes from 240. 1 for rounded answer.
3.	$\dfrac{AC}{12} = \dfrac{15}{9}$, so $AC = 20$ cm. $BC = 20 - 12 =$ **8 cm**.	2	1 for a correct scale-factor equation. 1 for BC, not AC.
4.(a)	$V = \pi(6)^2(20) = 720\pi$ $\mathbf{2.26 \times 10^3 cm^3}$	1	Accept $720\pi\text{cm}^3$.
(b)	$V = \dfrac{4}{3}\pi(3)^3 = 36\pi$ **113 cm³**	1	Accept $36\pi\text{cm}^3$.
(c)	$\text{Available volume} = 720\pi - 1800$ $\text{Number} = \dfrac{720\pi - 1800}{36\pi} = 4.0845...$ **4 spheres** can be added.	2	1 for the correct capacity calculation. 1 for rounding down to 4. Do not round volumes before dividing.
5.	$23.5 \le L < 24.5$, $0.795 \le p < 0.805$ Fewest panels: $\left\lceil \dfrac{23.5}{0.805} \right\rceil = 30$. Most panels: $\left\lceil \dfrac{24.5}{0.795} \right\rceil = 31$. Costs: \$975.00 and \$1007.50. Greatest difference: $(31 - 30) \times 32.50 =$ **\$32.50**.	4	1 for all length bounds. 1 for minimum 30 panels. 1 for maximum 31 panels. 1 for \$32.50. The limiting ratios give attainable integer counts despite the open upper bounds.

Coordinate Geometry • 11 marks

Question	Working and answer	Marks	Comments
1.(a)	-2	1	Correct gradient.
(b)	$4 - 2(15) = -26 \neq -24$. **No**, the point is not on L.	2	1 for substitution and value -26. 1 for conclusion.
2.(a)	$\left(\dfrac{-2+6}{2}, \dfrac{7+1}{2}\right) = (2, 4)$	1	Correct midpoint.
(b)	$AB = \sqrt{(6+2)^2 + (1-7)^2}$ $= \sqrt{64 + 36} = 10$ units.	2	1 for correct distance expression. 1 for 10.
(c)	$m_{AB} = \dfrac{1-7}{6+2} = -\dfrac{3}{4}$ $m_{\text{perpendicular}} = \dfrac{4}{3}$ $y - 4 = \dfrac{4}{3}(x - 2)$ $3y - 12 = 4x - 8$ $\mathbf{4x - 3y = -4}$	5	1 for gradient of AB. 1 for perpendicular gradient. 1 for line through midpoint $(2, 4)$. 1 for correct rearrangement. 1 for the required integer form. Accept non-zero integer multiples.

Graphs of Quadratic Functions • 24 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.(a)	For $x = -2, -1, 0, 1, 2, 3$: $y = 2, -2, -4, -4, -2, 2$.	2	1 for at least three correct values. 2 for all six correct.
(b)		3	1 for correct plotted points. 1 for domain $-2 \le x \le 3$ only. 1 for a smooth parabola. The full grid is shown; the curve must stop at $(-2, 2)$ and $(3, 2)$.
2.(a)	$f(x) = (x + 2)^2 - 4$ Vertex $(-2, -4)$; x-intercepts $(-4, 0), (0, 0)$.	3	1 for vertex. 1 for both intercepts. 1 for upward parabola with correct shape and labels.
(b)	$g(x) = (x - 1)^2 - 1$ Vertex $(1, -1)$; x-intercepts $(0, 0), (2, 0)$.	3	1 for vertex. 1 for a second correct point or both intercepts. 1 for correct curve and function label.
(a), (b)	 Left vertex: f. Right vertex: g.	—	Reference sketches. Student sketches need not be to scale, but intercepts and vertices must agree with the formulae.
(c)	A **translation** 3 units to the right and 3 units up.	2	1 for each component with correct direction. The answer must identify a translation.

Graphs of Quadratic Functions continued			
Question	**Working and answer**	**Marks**	**Comments**
3.(a)	$h(0) = 1 + 4(0) - 0^2 = 1$ **1 m**	2	1 for substituting $t = 0$. 1 for answer.
(b)	$h(2) = 1 + 4(2) - 2^2 = 5$ **5 m**	2	1 for substitution. 1 for answer.
(c)	$0 = 1 + 4t - t^2$ $t^2 - 4t - 1 = 0$ $t = \dfrac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}$ $t = -0.236...$ or $t = 4.236...$ **4.24 seconds**. The negative time is before launch and outside the model.	3	1 for the correct quadratic. 1 for solving it. 1 for the positive time only, with a reason for rejecting the negative time.
(d)	$2 = 1 + 4t - t^2$ $t^2 - 4t + 1 = 0$ $t = \dfrac{4 \pm \sqrt{16 - 4}}{2} = 2 \pm \sqrt{3}$ **0.268 s and 3.73 s**. Both are after launch and before $2 + \sqrt{5}$ seconds.	4	1 for setting $h = 2$. 1 for rearrangement. 1 for solving. 1 for both valid times. Neither root may be omitted.

Geometry • 14 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.	$\angle ABC = 180° - 128° = 52°$ $\angle ACB = 52°$ $e = 180° - 52° - 52° =$ **76°**.	4	1 for interior angle 52°. 1 for straight-line reason. 1 for 76°. 1 for both remaining reasons: equal base angles in an isosceles triangle and triangle angle sum 180°.
2.(a)	$m = 180° - 110° =$ **70°**.	2	1 for 70°. 1 for opposite angles of a cyclic quadrilateral summing to 180°.
(b)	$\angle ACB = 180° - 110° - 35° = 35°$ $j = 180° - 35° =$ **145°**.	4	1 for 35°. 1 for triangle angle sum. 1 for 145°. 1 for angles on a straight line. Accept a correct alternative method.
3.(a)(i)	$\overrightarrow{OM} = \boldsymbol{p} + \dfrac{1}{2}\boldsymbol{q}$	1	Correct expression.
(ii)	$\overrightarrow{ON} = \dfrac{1}{2}\boldsymbol{p} + \boldsymbol{q}$	1	Correct expression.
(b)	$\overrightarrow{MN} = \overrightarrow{ON} - \overrightarrow{OM}$ $= \dfrac{1}{2}(\boldsymbol{q} - \boldsymbol{p})$ $\overrightarrow{AC} = \boldsymbol{q} - \boldsymbol{p}$ Thus $\overrightarrow{MN} = \dfrac{1}{2}\overrightarrow{AC}$, so $MN \parallel AC$.	2	1 for $\overrightarrow{MN}$. 1 for comparison with $\overrightarrow{AC}$ and a valid parallel conclusion.

Total: 135 marks